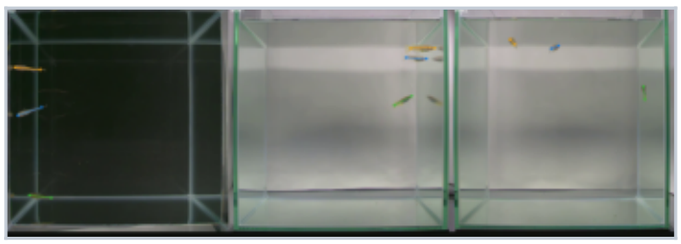
Tracking individual fish identities in 3D

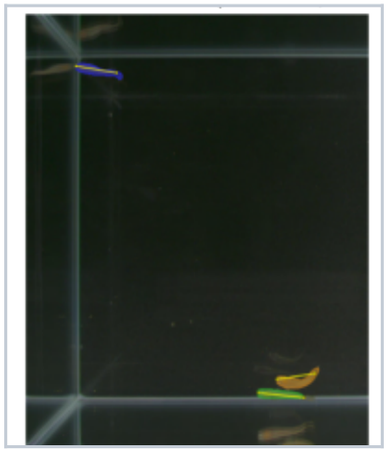
Multi-camera medaka mating pipeline: pose + segmentation → geometry → identity-stable 3D tracks

1 Multi-view capture



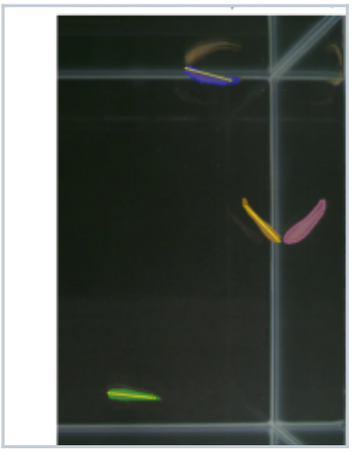
3 calibrated cameras (1 top + 2 side),
166 fps, 1F + 2M medaka in a 20 cm tank

2 SLEAP pose estimation



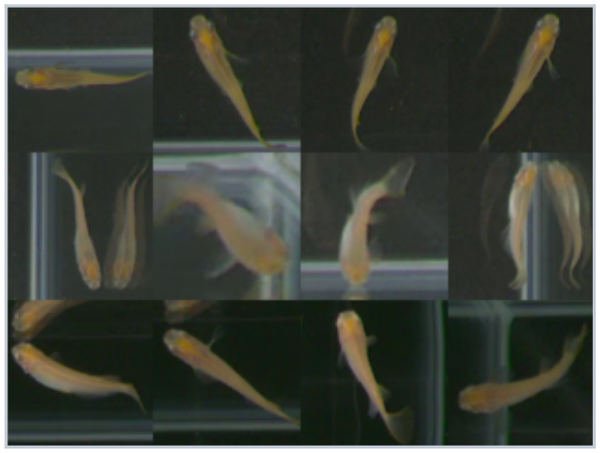
2-node skeleton (mouth→tail) per fish,
per view (yellow overlay)

3 Detectron2 segmentation



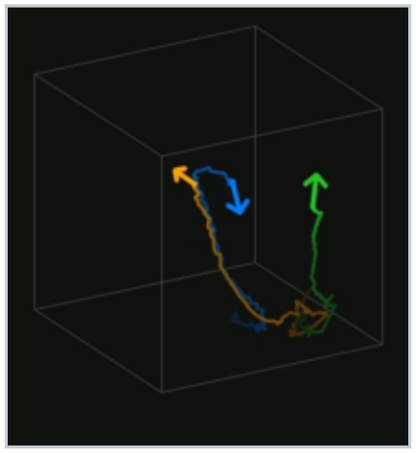
Per-instance masks gate detections and
feed appearance crops (filled colors)

4 Appearance ID



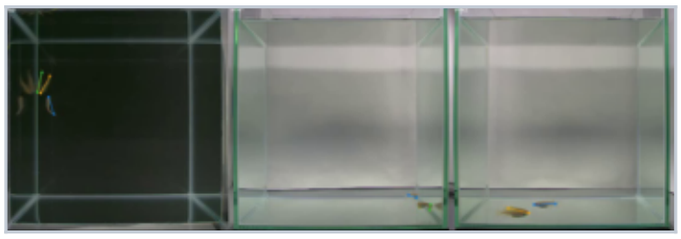
Per-fish mask crops → embedding;
the 3 fish separate (sil 0.3–0.4)

5 3D reconstruction + ID link



Cross-view triangulation via calibration;
NN + optic-flow + velocity/gap-aware
Hungarian linking unifies IDs in 3D

6 Stable ID through crossings



IDs held through close contact
(min inter-fish dist 2.3 mm)